WSD

WORD SENSE DISAMBIGUATION

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THE PURPOSE OF THIS PROJECT

It tests how different context sizes affect disambiguation quality and sentiment score using Ollama and Open English WordNet.

- assign correct WordNet senses to lemmas in a JSON corpus
- add sentiment scores for each concept
- assign a meaning tag to each concept
- evaluate the impact of context size on accuracy
- compare the model's predictions with human annotation

VIRTUAL ENVIRONMENT

```
PS C:\Users\Admin> ssh sone@158.194.46.11
sone@158.194.46.11's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is enabled.

16 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Last login: Wed Mar  4 20:50:33 2026 from 90.177.131.201
sone@nlplab:~$ cd JSP3/tasks/wsd-c
sone@nlplab:~/JSP3/tasks/wsd-c$  source .venv/bin/activate
(wsd-c) sone@nlplab:~/JSP3/tasks/wsd-c$ git pull
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 5 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.14 KiB | 582.00 KiB/s, done.
From https://github.com/bond-lab/JSP3
   0499604..b7f8cfe  main       -> origin/main
Updating 0499604..b7f8cfe
Fast-forward
 tasks/wsd-c/new_semproject.py | 20 +++++++++++++-------
 1 file changed, 13 insertions(+), 7 deletions(-)
(wsd-c) sone@nlplab:~/JSP3/tasks/wsd-c$  python new_semproject.py
Error: listen tcp 127.0.0.1:11434: bind: address already in use
Cached file found: /home/sone/.wn_data/downloads/0f5371187dcfe7e05f2a93ab85b4e1168859a5c2
Skipping oewn:2025+ (Open English Wordnet); already added
```

INPUT DATA

twwtn-en_human (1).json
JSON corpus with:

- Sentences (sent)
- Concepts per sentence (conc)

Each concept contains:

- Lemma (clemma)
- Gold tag (for evaluation)

```
{
  "sent": {...},
  "conc": {
      "sentence_id": {
          "concept_id": {
              "clemma": "...",
              "tag": "..."
          }
      }
  }              # example structure
}
```

HOW DOES IT WORK

1. Start Ollama server
2. Load WordNet synsets
3. Extract sentences and concepts
4. Build a contextual prompt
5. LLM selects:
- WordNet synset OR
- Special tag (per, loc, org, x, w...)
6. Assign lexical sentiment score
7. Save tagged and stats JSON
8. Match between predicted tag and gold tag

- Main corpus: twwtn-en_human (1).json
- Processing: 50 sentences
- Context sizes: 0–3

PROBLEMS

Context size 0 Correct: 75, Total: 1404, Accuracy: 4.86
Context size 1 Correct: 66, Total: 1390, Accuracy: 4.58
Context size 2 Correct: 65, Total: 1384, Accuracy: 4.7
Context size 3 Correct: 66, Total: 1404, Accuracy: 4.53

I didn't use evaluation code from github..

I needed to write better prompt.


```
"4": {
        "clemma": "second",
        "tag": "15235126-n",
        "wids": [
          5
        ]
      },
      "5": {
        "clemma": "a",
        "tag": "x",
        "wids": [
          6
        ]
      },
      "6": {
        "clemma": "brown",
        "tag": "00372111-a",
        "wids": [
          7
        ]
      },
```

```
"13": {
    "clemma": "raise",
    "tag": null,
    "wids": [
      18
    ]
  },
  "14": {
    "clemma": "his",
    "tag": null,
    "wids": [
      19
    ]
  },
  "15": {
    "clemma": "eyebrow",
    "tag": null,
    "wids": [
      20
    ]
  },
```

RESULTS

Model	ma 3 (Acc withou	wen 2.5 (Acc without'	emma 2 (Acc without'	Llama 3 (Macro F1)	Qwen 2.5 (Macro F1)	Gemma 2 (Macro F1)
Context size 0	6.05%	6.31%	4.50%	0.173	0.4	0.386
Context size 1	5.92%	5.66%	4.25%	0.13	0.347	0.412
Context size 2	5.79%	5.41%	4.12%	0.123	0.369	0.416
Context size 3	5.79%	5.66%	3.86%	0.114	0.33	0.406

FUTURE WORK

- Run 100 sentences
- Test more models (especially mistral-nemo:12b!!)
- Try to improve the code
- Rewrite the prompt

THANK YOU FOR YOUR ATTENTION!